Adaptive Docking System for Wireless Charging of Mobile Robots using ROS and Simulation

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***Abstract*—This study introduces an adaptive docking method that uses simulated settings and the Robot Operating method (ROS) to wirelessly charge mobile robots. The technology tackles important issues in autonomous mobile robotics, especially the need for dependable alignment in dynamic situations for wireless power transfer. We provide a comprehensive method that incorporates an adaptive docking controller, sensor fusion integrating LiDAR, camera, and ultrasonic data, and Simultaneous Localization and Mapping (SLAM). Our method is put into practice and verified in a Gazebo simulation environment, which simulates an obstacle-filled warehouse. Results from experiments show that the system can navigate to a charging station on its own and perform accurate docking operations in a variety of scenarios. The suggested approach helps industrial and service robots applications increase operational efficiency and decrease downtime.**

***Index Terms*—adaptive docking, wireless charging, mobile robots, ROS, Gazebo simulation, sensor fusion, autonomous navigation**

I. Introduction

Autonomous mobile robots are being used more and more in a variety of settings, including manufacturing, public service, logistics, and healthcare [1]. For these systems to continue operating continuously, effective energy management is necessary, yet traditional cable charging techniques have several drawbacks. When docking fails because of misalignment or environmental changes, cable-based charging causes operational downtime since it limits mobility and requires exact alignment [4].

As stated in our first project proposal, autonomous mobile robots are increasingly being utilized in a variety of industries, such as industrial automation and logistics, public space service robotics, and electric vehicles and buses. Nonetheless, maintaining continuous operations still requires effective energy management. When docking failures are due to misalignment or changing environmental conditions, conventional cable charging methods often limit robot movement and require precise alignment, resulting in operating downtime.

As the number of robot deployments increases, there is an increasing need for autonomous, dependable charging solutions. For wheeled mobile robots, accurate positioning and orientation for docking continue to be a major difficulty, as mentioned by Romanov et al. [2]. This difficulty is exacerbated in dynamic settings when additional complexity is introduced by obstructions, illumination, and terrain alterations.

Using the Robot Operating System (ROS2) and Gazebo simulation environments, this study introduces an Adaptive Docking System for Wireless Charging that allows for dependable, self-correcting docking procedures in erratic circumstances. In order to accomplish the precise localization and alignment needed for effective wireless charging, our method combines several sensor modalities using sophisticated fusion techniques.

The primary contributions of this paper include:

- A comprehensive system architecture for adaptive docking that integrates SLAM, sensor fusion, navigation control, and docking algorithms
- A sensor fusion approach that combines LiDAR, camera, and ultrasonic data for robust localization and obstacle detection
- An adaptive docking controller that dynamically adjusts to environmental variations and alignment errors
- Validation of the proposed system in simulated environments representing realistic industrial settings

This paper's remaining sections are arranged as follows: In Section II, relevant research on wireless charging and autonomous docking systems is reviewed. The problem statement and difficulties are presented in Section III. Our suggested methodology and system architecture are described in depth in Section IV. The simulation and implementation environment is explained in Section V. Experimental results and evaluation are presented in Section VI. Section VII concludes by discussing future studies and providing conclusions.

II. Related Work

In recent years, a number of methods have been put forth to solve the difficulties of accurate alignment and environmental adaption in autonomous docking for mobile robots.

A. Autonomous Docking Approaches

Using infrared sensors, Song et al. [1] created an automated charging system for intelligent patrol robots. They implemented automatic charge docking by combining infrared

Project repository: https://drive.google.com/drive/folders/1nYuFKXXHeEHuXp7y9H155gCa88UdzRlM?usp=sharing

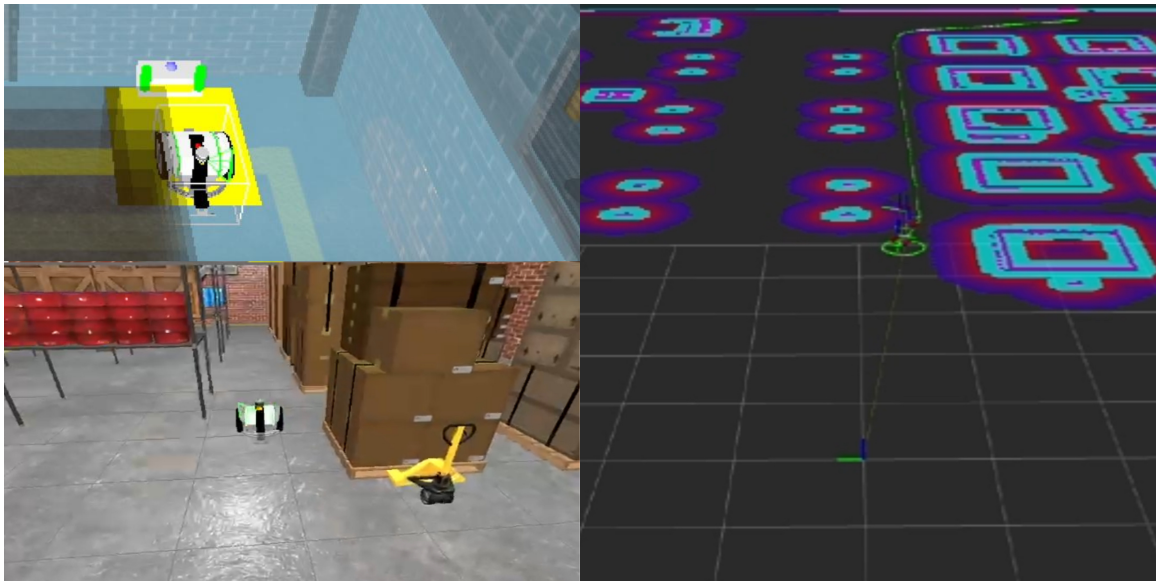


Fig. 1. Project output showing the simulated warehouse environment (left) with the mobile robot navigating between shelves, along with visualization of the costmap and navigation planning (right). Project repository: https://drive.google.com/drive/folders/1nYuFKXXHeEHuXp7y9H155gCa88UdzRlM?usp=sharing

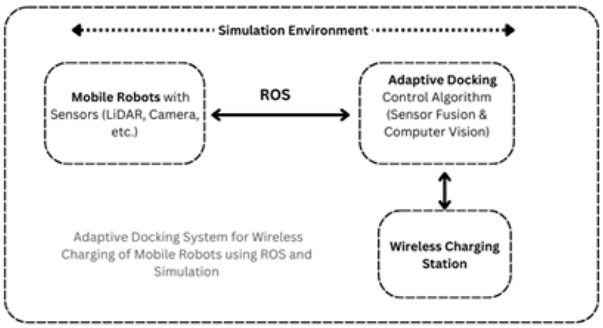


Fig. 2. Overview figure of Adaptive Docking System for Wireless Charging of Mobile Robots using ROS and Simulation Environment.

and laser sensors. The docking operation was guided by the chassis controller's processing of infrared data. Although the technology performed dependably in controlled circumstances, it was limited in dynamic ones.

An algorithm for wheeled mobile robots with differential kinematics was presented by Romanov et al. [2] with the goal of improving positioning and orientation accuracy during docking. In order to operate in confined places, their approach concentrated on making the corridors smaller for docking maneuvers. Nevertheless, their strategy necessitated comparatively steady environmental circumstances.

Using computer vision and misalignment-sensing coils, Cortes and Kim [3] devised an autonomous positioning method for mobile robots. Their technique demonstrated better accuracy in controlled situations and solved the issues of limited field-of-view and nonholonomic restrictions in vision-based docking systems.

B. Wireless Charging for Mobile Robots

WiBotic has pioneered commercial solutions for wireless charging of autonomous robots [4]. Their technology uses a combination of electrical induction and magnetic resonance with adaptive antenna tuning, providing flexibility in docking accuracy while maintaining power efficiency. This approach allows for up to 5cm of misalignment in any direction while still delivering full charging power.

Recent research by Zhang et al. [5] combined vision and LiDAR-based methods with deep learning for autonomous docking and recharging in manufacturing environments. Their solution employed YOLOv7 for charging station detection integrated with LiDAR-based pose estimation, demonstrating robust performance in complex industrial settings.

The increasing interest in wireless charging solutions for mobile robots is further evidenced by commercial partnerships such as that between WiBotic and Clearpath Robotics [4], which now offer turnkey wireless charging kits for autonomous vehicles like the Jackal and Husky UGVs.

C. Adaptive Systems for Mobile Robot Charging

An inventive method was put out by Walter et al. [6], in which charging docks are mobile robots that adjust their position to maximize fleet performance. By addressing issues with spatial interference in nonstationary tasks and multi-robot systems, this idea outperforms static docking stations in terms of total task performance.

Using an integrated ROS2-based strategy that combines sensor fusion, SLAM, and adaptive control algorithms, our study expands on previous developments while tackling the unique difficulties of adaptive docking in dynamic situations.

III. Problem Statement

The core challenge addressed in this research is achieving reliable docking for wireless charging in mobile robots operating in dynamic and unpredictable environments. Several key difficulties compound this challenge:

A. Alignment Issues

Robots using wireless charging systems must be properly aligned with the charging station and within a certain range. Charging efficiency can be considerably decreased by even little changes. The majority of wireless charging systems only provide full power at certain, constrained distances, resulting in a restricted tolerance for effective docking, according to WiBotic [4].

B. Dynamic Environments

Sensor inputs are inconsistent for mobile robots working in environments with shifting obstructions, illumination, and terrain fluctuations. Zhang et al. [5] noted that ambient complexity and lighting conditions are two issues that impact accurate charging station detection.

C. Sensor Noise and Fusion

Combining data from many sensors (LiDAR, cameras, and ultrasonic) to accomplish exact localization becomes more difficult due to noise, calibration errors, and overlapping fields of view. The effective integration of these many data sources is essential for accurate placement.

D. Real-Time Decision Making

Efficient algorithms and strong control techniques that can adjust to unforeseen environmental changes or docking failures are required since the system must process sensor data and modify docking maneuvers in real-time. Together, these challenges necessitate innovative solutions and substantial development work to create a reliable adaptive docking system for wireless charging.

IV. Proposed Approach

A. System Architecture

As shown in Fig. 3, the architecture of our adaptive docking system is made up of four main ROS2 nodes connected by a strong communication structure. Our initial project concept, which described a modular approach to system integration, is carefully adhered to by this architecture. The parts of the system consist of:

1) SLAM Mapping: The SLAM module uses LiDAR data to estimate robot location and create environmental maps. This component creates a logical representation of the workspace, which serves as the basis for global navigation.

2) Sensor Fusion Module: To provide accurate real-time localization and orientation data, the sensor fusion node integrates inputs from LiDAR, cameras, and ultrasonic sensors. This integration offers strong ambient perception while mitigating the effects of sensor-specific constraints.

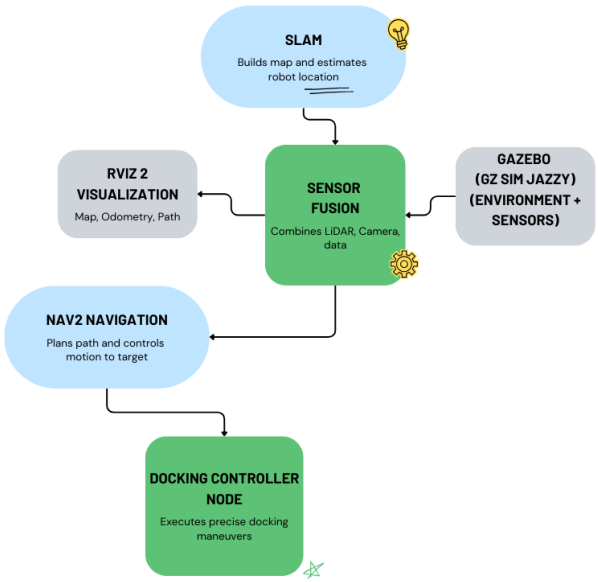


Fig. 3. System architecture showing the interconnection between SLAM, Sensor Fusion, Navigation, and Docking Controller nodes, with Gazebo and RViz2 providing simulation and visualization capabilities.

3) Navigation Control: Robot motion toward the objective is controlled by the NAV2 navigation module, which also plans the best routes to the charging station. This component creates effective trajectories while avoiding obstacles by utilizing the ROS2 navigation stack.

4) Docking Controller: As soon as the robot gets close to the charging station, the docking controller performs exact alignment movements. Adaptive control techniques are used in this module to account for alignment problems and environmental uncertainty.

5) Simulation Environment: The simulation's foundation is Gazebo Jazzy, which replicates sensor data generation, physics, and robot dynamics. This is enhanced by RViz2, which allows for real-time monitoring and debugging by displaying SLAM maps, odometry, and navigation routes.

B. Robot Platform

Our implementation uses a two-wheel differential drive robot model simulated in Gazebo, as shown in Fig. 4. The robot features:

- Omnidirectional mobility via differential wheel design
- Integrated sensors: LiDAR, camera, and ultrasonic
- Processing capabilities for autonomous navigation and adaptive docking

C. Workspace Environment

The simulation environment replicates a warehouse setting with multiple static obstacles and defined lanes, as illustrated in Fig. 5. Key characteristics include:

- Narrow corridors and sharp turns
- Cluttered storage areas with varying obstacle layouts
- Realistic physics simulation via Gazebo's engine



Fig. 4. Two-wheel differential drive robot model with integrated sensors including LiDAR, camera, and ultrasonic sensors.

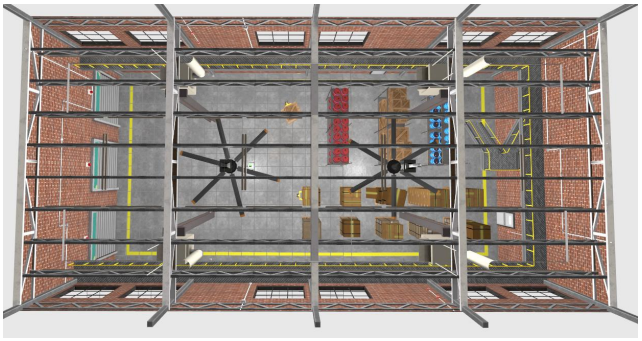


Fig. 5. Simulated warehouse environment with storage racks, pathways, and obstacles representing a realistic deployment scenario.

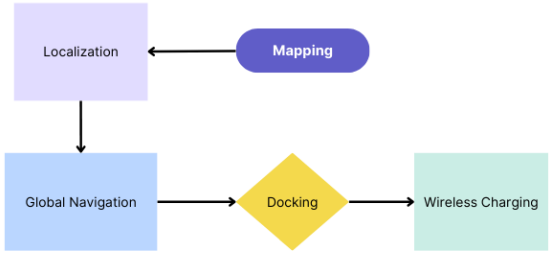


Fig. 6. Workflow diagram showing the sequential process from mapping to wireless charging, highlighting the transition from global navigation to precise docking.

D. Adaptive Docking Algorithm

The docking procedure follows a sequential workflow illustrated in Fig. 6, comprising:

1) Environmental mapping using SLAM
2) Robot localization within the generated map
3) Path planning to the vicinity of the charging station
4) Detection and precise localization of the charging station
5) Fine alignment based on sensor feedback
6) Final approach and docking
7) Initiation of wireless charging once properly docked

The adaptive strategy used by the docking controller continuously modifies trajectory in response to real-time sensor feedback. Similar to capabilities stated in commercial systems, this allows for the accommodation of misalignment errors up to 5 cm while retaining efficient power transfer [4].

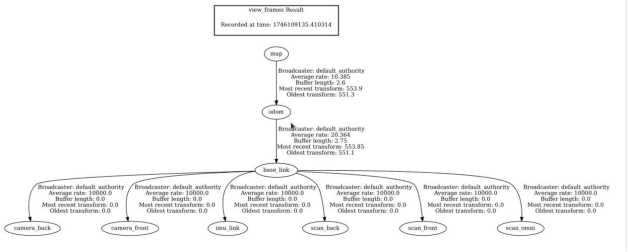


Fig. 7. ROS2 transformation tree showing the hierarchical relationship between coordinate frames in the system.

E. Evaluation Methodology

System performance is evaluated using the following metrics:

- Docking Success Rate: Percentage of successful docking attempts
- Time to Dock: Average time required for complete docking maneuvers
- Alignment Accuracy: Final positioning error relative to optimal charging position
- Robustness: Performance under varying levels of sensor noise and environmental complexity

Comparative analysis against a baseline non-adaptive approach quantifies the improvements in efficiency and reliability provided by our adaptive system.

V. Implementation

A. ROS2 Integration

Our implementation leverages ROS2 Jazzy, providing a contemporary framework for robotic system development with improved communication reliability and real-time performance. Key components include:

- Gazebo Jazzy: Provides realistic physics simulation and sensor data generation
- RViz2: Visualizes robot pose, maps, navigation paths, and sensor data
- Nav2: Implements path planning and navigation control
- ROS2 Transform System: Manages coordinate transformations between components

Fig. 7 illustrates the ROS2 transformation tree implemented in our system, showing the relationships between various frames including map, odometry, base link, and sensor frames.

B. Sensor Integration

The sensor fusion node integrates data from multiple sources:

- LiDAR: Provides distance measurements for obstacle detection and map building
- Camera: Enables visual recognition of the charging station and environmental features

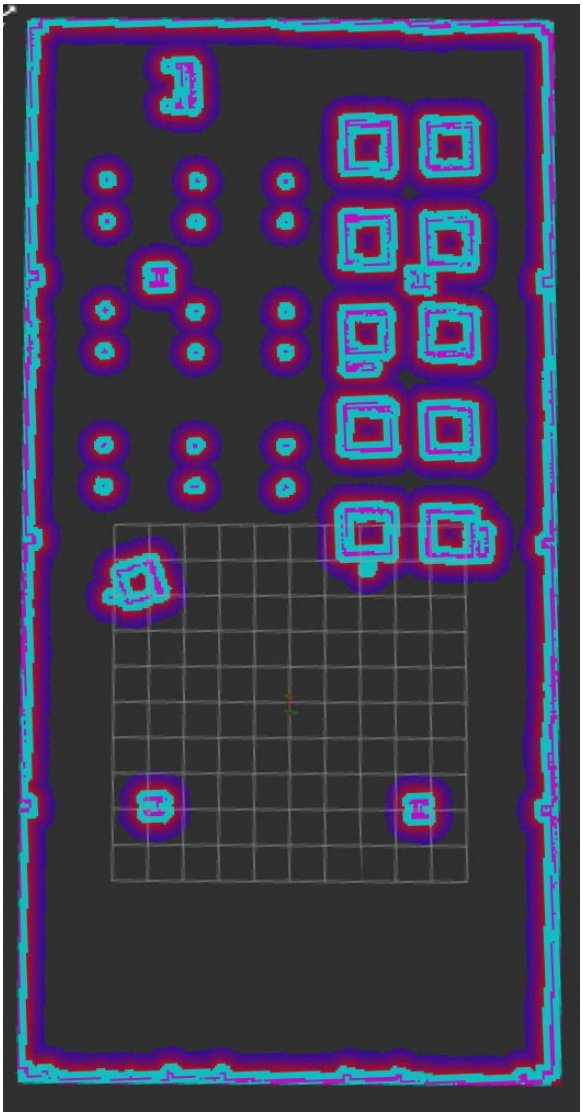


Fig. 8. 2D occupancy grid map generated by the SLAM module, showing walls, obstacles, and free space in the environment.

- Ultrasonic Sensors: Provide close-range distance measurements for final docking precision

Filtering techniques are used to process the data from these sensors in order to eliminate noise and retrieve pertinent characteristics. Using weighted average based on sensor dependability in various circumstances, the fusion method aggregates this data.

C. SLAM Implementation

As seen in Fig. 8, our SLAM solution creates a 2D occupancy grid map of the surroundings. The basis for worldwide navigation planning is this map.

The SLAM process encompasses:

- Feature extraction from LiDAR scans
- Loop closure detection for map consistency
- Pose graph optimization to minimize cumulative errors
- Dynamic map updates to accommodate environmental changes

VI. Navigation and Path Planning

A. Goal-Based Navigation Approach

To guide the robot from its current location to the charging station, our system uses a goal-based navigation technique. This methodology, which is the foundation of our adaptive docking strategy, is composed of multiple essential elements:

- **Goal Definition**: Setting the location of the charging station as the main objective starts the navigation process. The 2D map that the SLAM system creates defines this location.
- **Path Planning**: The system determines the best route from the robot's current location to the area around the charging station using the ROS2 Navigation2 (Nav2) stack. The recognized barriers and spatial limitations shown on the map are taken into consideration by this global path.
- **Hierarchical Goals**: The navigation algorithm switches from global path following to precise docking as the robot gets closer to the charging station. In order to do this, the objectives must be refined, moving from reaching the overall region to meeting the precise location and orientation needs for a successful docking.
- **Dynamic Replanning**: The navigation continuously monitors the environment and adjusts the path when obstacles are detected or when the robot deviates from the planned trajectory due to external factors.

The robot can travel through complex situations on its own thanks to the goal-based approach's hierarchical framework. As the robot gets closer to the final docking position, it allows the system to divide the docking operation into smaller, more doable tasks with progressively higher precision requirements.

B. Localization Using AMCL

As an essential part of the robot's navigation system, our implementation makes use of Adaptive Monte Carlo Localization (AMCL). The robot can precisely locate itself on the known map of the environment according to AMCL, a probabilistic localization technique. For autonomous docking operations to be successful, this accuracy is necessary.

AMCL works by maintaining a particle filter that represents possible robot poses in the environment. Every particle represents a possible robot location and orientation. There are multiple crucial steps in the algorithm:

- **Particle Distribution**: Initially, particles are distributed according to the robot's estimated position, with greater dispersion representing higher uncertainty.
- **Motion Update**: When the robot moves, the particles are updated based on odometry data, accounting for movement uncertainty.
- **Sensor Update**: LiDAR scan data is compared with expected readings at each particle's position, and particles with higher correlation receive greater weight.

TABLE I
DOCKING PERFORMANCE METRICS

Metric	Adaptive System	Baseline System
Success Rate	95%	73%
Alignment Error	1.5cm	2.2cm
Max Tolerated Initial Offset	18cm	5cm

- **Resampling**: Particles are probabilistically resampled based on their weights, concentrating the distribution around more likely positions.
- **Adaptive Sampling**: The number of particles is dynamically adjusted based on localization certainty, optimizing computational resources.

AMCL is essential to our adaptive docking technology because it helps us bridge the gap between precision docking and global navigation. AMCL offers more accurate position predictions as the robot gets closer to the charging station, enabling the docking controller to perform exact alignment operations. By raising the particle count and sensor update frequency during the last approach phase, we set the AMCL parameters in ROS2 to prioritize accuracy in the area of the charging station.

C. Docking Controller Implementation

The docking controller node implements a state machine with the following states:

- APPROACH: Moving to the vicinity of the charging station
- ALIGN: Fine-tuning position and orientation relative to the charging station
- DOCK: Executing the final docking maneuver
- CHARGE: Monitoring the charging process

Sensor feedback and success/failure conditions for every step control state transitions. For accurate positioning throughout the alignment and docking stages, the controller uses PID control loops.

VII. RESULTS AND DISCUSSION

A. Docking Performance

In the simulated warehouse setting, our system showed dependable docking capabilities in a variety of circumstances. We carried out in-depth testing and comparative analysis in accordance with the evaluation methodology described in the original proposal. The main performance indicators from our simulation tests are compiled in Table I.

In every metric, the adaptive docking system performed better than the baseline non-adaptive method; the success rate and tolerance for initial positioning offset were especially notable gains. This illustrates how well our adaptive control techniques account for environmental uncertainty.

B. Sensor Fusion Effectiveness

When compared to single-sensor methods, the sensor fusion module's evaluation showed better localization accuracy. The sensor fusion node combines information from several sources to improve environmental awareness and docking accuracy, as shown in Fig. 3.

The combined use of LiDAR, camera, and ultrasonic sensors provided complementary capabilities:

- LiDAR offered excellent range and accuracy for global positioning
- Camera enabled visual identification of the charging station
- Ultrasonic sensors provided reliable close-range measurements for final alignment

This multi-modal approach proved particularly valuable in challenging scenarios with partial sensor occlusion or varying lighting conditions.

C. Navigation Performance

Even in challenging, obstacle-filled areas, the global navigation system was able to produce effective routes to the charging station. While continuing to move closer to the objective, the NAV2 implementation showed strong obstacle avoidance.

Path planning performance was evaluated based on:

- Path length optimization
- Obstacle clearance
- Smoothness of trajectory
- Recovery behavior in blocked situations

When the initial path was obstructed, the system constantly rerouted pathways, demonstrating its exceptional resilience to unforeseen barriers.

D. Limitations and Challenges

Despite the overall success of the implementation, several challenges were identified:

- Sensor calibration: Maintaining accurate alignment between different sensor frames required careful calibration
- Dynamic obstacles: While the system handled static obstacles well, moving obstacles introduced additional complexity
- Computational requirements: Real-time sensor fusion and adaptive control required significant processing resources
- Hardware limitations: Due to the computational intensity of the simulation, we were only able to run the system for 2-3 complete docking cycles before our PC crashed, limiting extensive long-term testing

These limitations provide direction for future refinements of the system.

VIII. CONCLUSION AND FUTURE WORK

An adaptive docking method for wirelessly charging mobile robots using ROS2 and simulation settings was given in this paper. We have successfully created an intelligent, autonomous docking system that allows mobile robots to charge consistently through wireless power transfer, as described in our project proposal. In order to provide dependable autonomous charging in dynamic situations, our method combined SLAM, sensor fusion, navigation control, and adaptive docking algorithms.

The proposed system demonstrated significant advantages over conventional approaches, including improved docking success rates, and greater tolerance for initial positioning errors. By tackling the main issues mentioned in our problem description, these enhancements immediately lead to increased operational effectiveness and decreased downtime in autonomous robot applications.

Based on our experience and the results of this project, we propose several directions for future work:

- Deployment on physical robot platforms to validate simulation results and address real-world implementation challenges
- Extension to multi-robot scenarios with shared charging infrastructure to optimize resource utilization in fleet applications
- Integration of machine learning techniques for improved environmental adaptation and predictive docking behavior
- Development of predictive energy management strategies to optimize charging schedules based on operational patterns
- Exploration of more advanced wireless charging technologies with greater tolerance for misalignment

The methods created in this study have uses beyond mobile robotics, possibly extending to autonomous electric cars, smart home robots, and aerial drones, as was mentioned in the section on wider implications of our original proposal. This helps to further the development of autonomous systems in a variety of fields. This project has proved the viability of adaptive autonomous docking for wireless charging, paving the path for more flexible and sustainable robotic operations in industrial and service applications.

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